CSA5108 Cryptography and Network security

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Exp 5: Affine Caesar Cipher
Code:

```
#include <stdio.h>
#include <ctype.h>

int gcd(int a, int b) {
   while (b != 0) {
      int t = b;
      b = a % b;
      a = t;
   }
   return a;
}

int modInverse(int a) {
   for (int i = 0; i < 26; i++) {
      if ((a * i) % 26 == 1)
         return i;
   }
   return -1;
}

int main() {
   char text[1000];
   int a, b, inv, choice;

   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);

   printf("Enter a: ");
   scanf("%d", &a);
```

```
    printf("Enter b: ");
    scanf("%d", &b);

    if (gcd(a, 26) != 1) {
        printf("Invalid value of a.\n");
        printf("a must be relatively prime to 26.\n");
        return 0;
    }

    inv = modInverse(a);

    printf("Allowed values of a: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23,
25\n");
    printf("There are no restrictions on b; b can be 0 through 25.\n");

    for (int i = 0; text[i] != '\0'; i++) {
        if (isupper(text[i]))
            text[i] = ((a * (text[i] - 'A') + b) % 26) + 'A';
        else if (islower(text[i]))
            text[i] = ((a * (text[i] - 'a') + b) % 26) + 'a';
    }

    printf("Ciphertext: %s", text);

    return 0;
}
```

